

VIP: VLM-Guided Interactive Policy Steering for Test-Time Preference Alignment

Shikun Ban, Lucky Kant Nayak, Shohei Nagai, and Zixi Song
Emails: {sban, lnayak, shoheina, zixis}@andrew.cmu.edu

Abstract—Human preferences in robot-assisted tasks are often nuanced, context-dependent, and difficult to specify in advance, making real-time preference alignment a central challenge for human–robot interaction. While Vision-Language Models (VLMs) can evaluate visual trajectories and infer likely user intent, they remain unreliable in ambiguous scenarios and lack awareness of their own uncertainty. We propose VLM-Guided Interactive Policy Steering (VIP), a test-time framework that combines VLM-based trajectory evaluation with confidence-aware human clarification. At each steering step, the system generates multiple candidate rollouts, uses a VLM to infer human-aligned preference values, and queries the user through visually grounded natural-language questions only when confidence is low. Our results demonstrate that (1) VLMs can generate effective clarifying queries grounded in visual trajectory differences, (2) natural-language QA achieves higher preference alignment with fewer interruptions than manual trajectory selection, (3) leveraging full interaction history significantly improves reasoning and reduces redundant queries, and (4) continuous user interaction throughout an episode yields better task performance than relying on a single initial instruction. Together, these findings show that interactive, confidence-driven VLM reasoning offers a scalable and efficient approach for aligning robotic behavior with human preferences at test time.

I. INTRODUCTION & MOTIVATION

Consider a robot assisting a person during bathing. A user may simply say “Please bathe me,” yet the correct behavior depends on subtle, individualized preferences: some may prefer firm wiping motions, others gentle strokes, and some may want the robot to avoid specific sensitive areas. Such distinctions are difficult to encode into reward functions or capture exhaustively through demonstrations [20, 10, 7], and can vary across users, environments, and even dynamically over time during policy execution. The core challenge is therefore not merely *performing* the task, but performing it in a way that *adapts* to nuanced, evolving human preferences at test time.

Existing approaches to preference alignment - for example RLHF - require large-scale human feedback or retraining, making them impractical for interactive, real-time scenarios. Language-based clarification through Large Language Models (LLMs) improves flexibility, yet language alone cannot reliably distinguish fine-grained differences in motion or trajectory style. Vision-Language Models (VLMs) address this gap by jointly interpreting images and text, enabling trajectory-level evaluation grounded in what the robot is physically doing rather than what the user can describe. However, VLM-based evaluation alone is often unreliable [9, 15]: when multiple trajectories appear similarly plausible, VLMs may express

high confidence in incorrect preferences or become biased by superficial cues. This suggests that VLMs are best used for *proposing* preferences and *identifying ambiguity*, while humans remain essential for resolving those ambiguities through well-targeted interaction.

This motivates an interactive preference alignment framework in which VLMs evaluate candidate trajectories, recognize when they are uncertain, and proactively query the user to refine their understanding.

Our Contribution. We introduce a confidence-based interactive VLM pipeline for test-time preference alignment that advances preference-guided robotic behavior along several dimensions: (1) we develop a VLM-based evaluator that not only scores candidate trajectories but also estimates its own uncertainty and, when unsure, generates visually grounded natural-language queries highlighting meaningful differences between trajectories, allowing humans to disambiguate subtle preferences far more effectively and efficiently than manual trajectory selection. (2) we demonstrate that natural-language QA increases and stabilizes confidence during inference compared to clarification through direct trajectory selection. (3) we show that leveraging full conversation history reduces redundant queries and improves alignment rates, suggesting that the VLM effectively extracts and integrates key information from the entire interaction history for the current scene. (4) Lastly, continuous user interaction throughout the episode yields better preference alignment than relying solely on an initial instruction.

II. RELATED WORK

A. Learning from Human Preferences

A central approach to aligning robot behavior with human intent is preference learning [3]. Methods like Reinforcement Learning from Human Feedback (RLHF) [2] learn a reward model from human preference comparisons and then optimize policies. To simplify this, methods such as Direct Preference Optimization (DPO) [11] fine-tune policies directly with a classification loss, and Inverse Preference Learning (IPL) [6] bypasses reward models by fitting Q-functions to preference data. While these methods highlight the promise of preference-based alignment, they typically assume fixed datasets or require fine-tuning, making them impractical for continuous, real-time adaptation to diverse and evolving user preferences in interactive Human-Robot Interaction (HRI) settings.

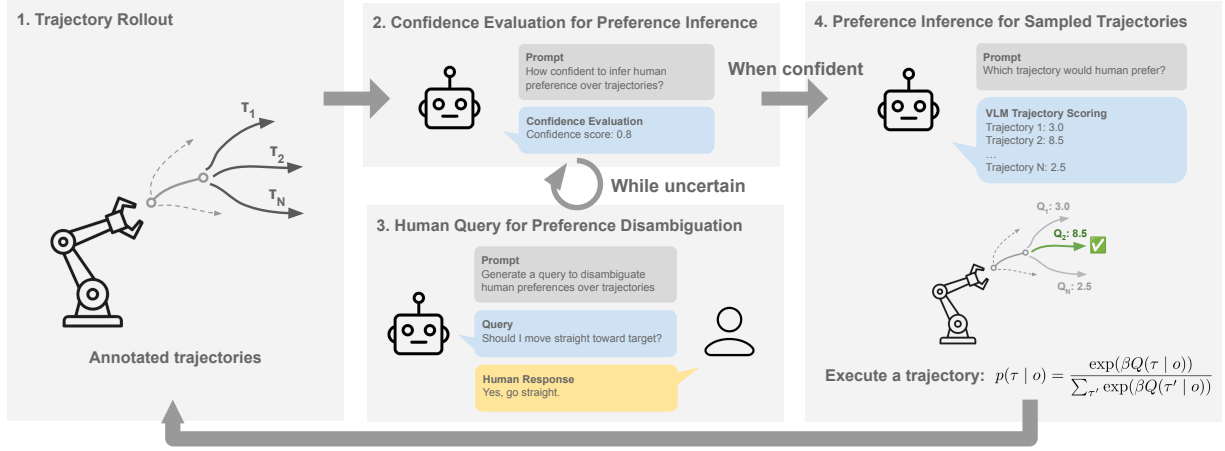


Fig. 1: VLM-Guided Interactive Policy Steering framework aligns robot behavior with human preferences by iteratively (1) generating candidate rollouts, (2) assessing confidence in inferred preferences, (3) engaging the human for clarification when ambiguity remains, and (4) inferring preference values for each trajectory and executing a trajectory via a Boltzmann distribution.

B. LLM and VLM for Preference Alignment

Recent advances have leveraged large foundation models as flexible proxies or critics for human preference. LLM-based approaches [5, 12, 14] enable robots to query, clarify, and update their policies from user input. Strong LLMs have been validated as effective human preference proxies, achieving high agreement with human judgments [21]. However, relying solely on language can be ambiguous, especially in tasks where behavior differences require low-level visual reasoning, not just high-level semantic distinction.

Vision-Language Models (VLMs) address this by extending capability to jointly reason over perception and language, making them powerful candidates for a visually grounded critic [18, 9, 19]. For instance, RL-VLM-F [18] and Preference VLM [4] use VLMs as critics for visual trajectory evaluation, and ROBOCLIP [16] uses VLMs for learning policies from single demonstrations. While VLMs offer richer context, prior work has highlighted their limitations in zero-shot reasoning about low-level physical actions [9]. Our work addresses the current limitations by proposing an interactive framework where we monitor the VLM’s confidence in preference inference and query human for disambiguating their preferences when uncertainty.

C. Test-Time Policy Steering

Policy steering has emerged as a lightweight alternative to fine-tuning [8, 19, 17]. V-GPS [8] enhances the performance of frozen generalist policies by re-ranking sampled actions according to a pre-trained value function. FOREWARN [19] employs a latent world model to predict action outcomes and aligns a VLM to reason over those latent states, enabling open-vocabulary verification of candidate actions and more robust mode selection at test time. Our work follows this policy steering formulation but addresses the ambiguity inherent in VLM-based evaluations in a more human-centered way. Rather than relying solely on a fixed reward function or direct VLM

outputs, we integrate VLM-based preference inference with a confidence-aware querying mechanism, enabling more robust and adaptive online steering.

III. PROBLEM FORMULATION

We consider an autonomous agent that aims to align its behavior with human preferences while performing a task. Since these preferences are latent and not explicitly defined a priori, the agent must infer them dynamically from observations and human feedback. We formalize this as a *test-time policy steering* problem [8], where human preference inference continuously guides policy execution at test time.

A. Preference Alignment

We define *preference alignment* as the process of selecting the trajectory that a human would most likely prefer, given the current state of the environment. The agent’s decision-making process is modeled via a Boltzmann distribution over possible trajectories [1]:

$$p(\tau | o) = \frac{\exp(\beta Q_h(\tau | o))}{\sum_{\tau'} \exp(\beta Q_h(\tau' | o))}, \quad (1)$$

where $Q_h(\tau | o)$ denotes the latent human value function over a trajectory τ conditioned on observation o , and $\beta > 0$ is the rationality coefficient that governs how deterministically choices are made.

The objective of preference alignment is to approximate the latent human value function Q_h with an estimator $\hat{Q}_h$. This enables the agent to adapt its behavior in a manner consistent with human preferences, thereby aligning its actions with the trajectories a human is most likely to select.

B. Interactive Policy Steering

Since directly evaluating all possible trajectories in continuous space is intractable, we adopt a sample-based steering approach. The agent uses a pre-trained base policy $\pi_{base}(a_t |$

o_t) to generate a discrete set of K candidate trajectories $\{\tau_1, \dots, \tau_K\}$ ($\tau_i = (o_t, a_t, \dots, o_{t+T})$), spanning a horizon of T steps.

For each trajectory τ_i , the agent estimates the human preference value $\hat{Q}_h(\tau_i | o_t)$ and normalizes these estimates into a Boltzmann distribution:

$$\hat{p}(\tau_i | o_t) = \frac{\exp(\beta \hat{Q}_h(\tau_i | o_t))}{\sum_{j=1}^K \exp(\beta \hat{Q}_h(\tau_j | o_t))}. \quad (2)$$

Standard steering methods would execute actions based solely on this distribution, assuming a reliable estimator $\hat{Q}_h$. However, $\hat{Q}_h$ may be uncertain—especially in states where multiple trajectories appear equally plausible from a human’s perspective.

To mitigate this, we introduce a *confidence-aware mechanism*. A scalar confidence metric $c(o) \in [0, 1]$ and a threshold c' determine whether the agent acts autonomously or queries the human. When confidence is low ($c(o) < c'$), the agent generates a *natural language query* to the human (e.g., “Should I go left or right?”). When confidence is high ($c(o) \geq c'$), the agent samples a trajectory from the distribution $\tau \sim \hat{p}(\tau_i | o_t)$.

This yields a unified framework of *interactive policy steering*, where the agent requests human feedback only when necessary, reducing user burden while maintaining preference alignment.

IV. PROPOSED APPROACH

In this section, we present the complete *VLM-guided interactive policy steering* framework (Figure 1). As outlined in Algorithm 1, the agent periodically generates and visualizes multiple candidate rollout trajectories from a base policy, leverage a VLM to evaluate human-alignment value for each candidate trajectory based on the visualization, and adaptively engages in clarification query with human when uncertainty remains high. The framework integrates three key components: (1) parallel rollouts and visual annotation, (2) confidence-aware human querying, and (3) preference inference and trajectory selection.

A. Parallel Rollout and Visual Annotation

Given a trained base policy $\pi(a_t | o_t)$, the agent periodically performs parallel rollouts to generate K candidate trajectories $\{\tau_i\}_{i=1}^K$ where each τ_i spans a horizon of T steps. To maintain temporal efficiency, rollouts occur every T control steps or within a maximum of M rollout cycles per episode, with $T = \lceil N/M \rceil$ for an episode of N steps. This corresponds to line 2 in Algorithm 1.

Each set of trajectories shares a common starting observation o_t and its corresponding image frame I_t . From each τ_i , we extract end-effector (EE) positions $\{p_{t:t+T}^{EE}\}$ and render them on I_t to form an annotated image $\hat{I}_t$ (Figure 2, left). Consecutive EE positions are connected by red line segments, and the final position is marked with a numbered red circle. In 2D manipulation tasks such as Push-T, these coordinates correspond to the planar control points (x_t, y_t) of the robot arm. This corresponds to line 3 in Algorithm 1.

Algorithm 1: VLM-Guided Interactive Policy Steering

Input: Base policy π , observation o , VLM

Notation: Image observation $I \in o$;

Candidate trajectories $\{\tau_i\}$; Interaction history $\mathcal{H}$;

Preference score $Q_{\text{VLM}}(\tau | \hat{I}, \mathcal{H})$;

Confidence score $C_{\text{VLM}}(\hat{I}, \mathcal{H})$; Confidence threshold c' ;

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1 while state is not terminal do
2    $\{\tau_i\}_{i=1}^K \leftarrow \text{Rollout}(\pi, o_t)$ ;
3    $\hat{I}_t \leftarrow \text{Annotate}(I_t, \{\tau_i\}_{i=1}^K)$ ;
4    $c_t \leftarrow C_{\text{VLM}}(\hat{I}_t, \mathcal{H}_t)$ ;
5   while  $c_t < c'$  do
6      $q_t \leftarrow \text{VLM.GenQuestion}(\{\tau_i\}, \hat{I}_t, \mathcal{H}_t)$ ;
7      $r_t \leftarrow \text{QueryHuman}(q_t)$ ;
8      $\mathcal{H} \leftarrow \mathcal{H} \cup \{(\hat{I}_t, q_t, r_t)\}$ ;
9      $c_t \leftarrow C_{\text{VLM}}(\hat{I}_t, \mathcal{H}_t)$ ;
10   $\hat{p}(\tau_i | \hat{I}_t, \mathcal{H}_t) \leftarrow \text{CalcProb}(\{Q_{\text{VLM}}(\tau_i | \hat{I}_t, \mathcal{H}_t)\}_{i=1}^K)$ ;
11   $\tau^* \sim \hat{p}(\tau | \hat{I}_t, \mathcal{H}_t)$ ;
12  Execute  $\tau^*$  and update  $o_t$ ;
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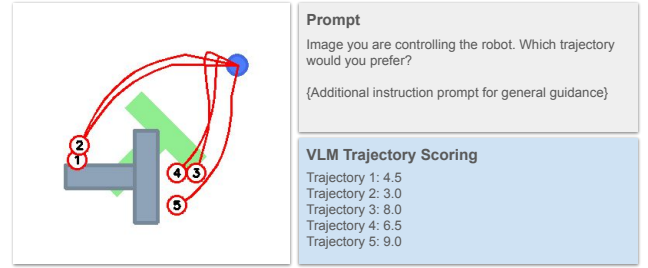


Fig. 2: Left: Annotated image (Push-T Task) with five candidate trajectories, each labeled with an index. Right (top): Partial prompt provided to the VLM, instructing it to infer human preferences and assign a score to each trajectory. Details in IV-C1. Right (bottom): VLM output showing trajectory scores in the range $[0, 10]$, representing the inferred preference value of each trajectory.

The annotated image $\hat{I}_t$ thus summarizes the K possible rollouts visually, serving as the shared context between the robot and the VLM-human system. The VLM uses $\hat{I}_t$ to infer trajectory preferences and confidence, while the human interacts only through natural language, never seeing the annotation overlays.

B. Confidence-Aware Human Querying

This component determines when to rely solely on the VLM for inference and when to involve the human through clarification. At each iteration, the VLM is prompted to assess its certainty regarding inferred trajectory preferences, and additional human input is requested only when necessary. At the beginning of each task, the VLM always starts by asking the human for a brief task description to establish shared context. This process corresponds to lines 4–9 of Algorithm 1.

1) *Confidence Evaluation via VLM*: Given the annotated image $\hat{I}_t$ and the interaction history $\mathcal{H}_t$, the system prompts the VLM to produce a normalized scalar confidence estimate [13] (line 4 of Algorithm 1):

$$c_t = C_{\text{VLM}}(\hat{I}_t, \mathcal{H}_t) \in [0, 1],$$

where $c_t = 1$ indicates complete certainty and $c_t = 0$ indicates maximum uncertainty. When $c_t \geq c'$, where c' is a predefined threshold, the system proceeds directly to preference inference without querying the human. Appendix A-A contains the full prompt specification for confidence evaluation.

2) *Querying Human When Confidence is Low*: If $c_t < c'$, the system enters a clarification phase (lines 5-9 of Algorithm 1) via a web-based user interface (UI). The VLM generates a natural language *query* q_t aimed at resolving the primary source of uncertainty, and the human responds with an answer r_t . The interaction history is then updated as:

$$\mathcal{H}_{t+1} = \mathcal{H}_t \cup \{(\hat{I}_t, q_t, r_t)\}.$$

To ensure natural and goal-oriented dialogue, the VLM is instructed not to reference any internal visual annotations (e.g., path labels or colors). Instead, it expresses uncertainty in semantic terms related to the task (e.g., “Should the robot push directly, or position itself first for a gentler approach?”). The Appendix A-C details the prompt used to generate clarifying questions. After incorporating the human response, the system re-evaluates confidence and repeats the process until $c_t \geq c'$.

C. Preference Inference and Trajectory Selection

Once sufficient confidence is achieved, the system prompts the VLM to infer human-aligned preference values for each candidate trajectory, conditioned on the annotated image and the evolving interaction history.

1) *Trajectory Scoring via VLM*: Using its visual reasoning capabilities, the VLM outputs a scalar preference estimate for each τ_i :

$$Q_{\text{VLM}}(\tau_i \mid \hat{I}_t, \mathcal{H}_t), \quad i = 1, \dots, K.$$

The prompt guiding this inference encourages structured reasoning about: (i) task intent, (ii) physical feasibility and unintended interference, (iii) alignment with the desired goal state, and (iv) the usefulness of the resulting configuration for subsequent actions. The complete prompt used for trajectory-level preference scoring is provided in Appendix A-B.

2) *Probability-Based Trajectory Selection*: The inferred preferences are converted into a selection distribution using:

$$p(\tau_i \mid \hat{I}_t, \mathcal{H}_t) = \frac{\exp(\beta Q_{\text{VLM}}(\tau_i \mid \hat{I}_t, \mathcal{H}_t))}{\sum_{j=1}^K \exp(\beta Q_{\text{VLM}}(\tau_j \mid \hat{I}_t, \mathcal{H}_t))},$$

where $\beta > 0$ controls the sharpness of the distribution (line 10 of Algorithm 1). The system either selects the highest-probability trajectory $\tau_t^* = \arg \max_{\tau_i} p(\tau_i \mid \hat{I}_t, \mathcal{H}_t)$, or samples according to $p(\tau \mid \hat{I}_t, \mathcal{H}_t)$ (line 11 of Algorithm 1). Executing τ_t^* advances the environment, producing updated observations that are appended to $\mathcal{H}$, enabling progressive refinement of both preference inference and confidence calibration.

V. RESULTS

In this section, we first outline our hypotheses (**H0–H4**) in V-A. We then present qualitative evaluation results for **H0** in V-B, quantitative evaluation results for **H1** and **H2** in V-C, and ablation results for **H3** and **H4** in V-D.

We choose Push-T as our primary evaluation task because it naturally contains clear differences in human preferences through multi-modal trajectory options. Its lightweight simulation and easy visualization further make it ideal for benchmarking preference alignment in policy steering.

A. Hypothesis

We list our hypotheses as follows:

(H0) Our VLM-based framework is able to generate visually grounded natural-language queries to human, effectively disambiguating human preferences.

(H1) VLM-based natural-language QA provides more effective and efficient preference alignment than manual trajectory selection.

(H2) VLM-based natural-language QA is better at improving and maintaining the VLM’s confidence about human preference than manual trajectory selection.

(H3) Full user conversation history leads to more effective and efficient preference alignment than partial user conversation history.

(H4) Continuous user-interaction of the VLM throughout the episode leads to better preference alignment than providing only initial task instruction to VLM.

B. Qualitative results of the VLM-generated natural-language query

The qualitative example shown in Figure 3 demonstrates that human preferences can be effectively disambiguated with the VLM-generated natural-language query. Given the example trajectory-annotated image, the VLM initially has low confidence, thereby querying human “Should I push it from the left side, or approach from the right side?”. This query is grounded in the actual visual trajectory options. After receiving the human answer “Please push the grey T shape from the left side.”, the VLM’s confidence increases and the outputted trajectory probability distribution favors Trajectory 1 and 2 on the left side of the T-block.

C. Quantitative comparison of VLM-based natural-language QA and manual trajectory selection

In our method, when the robot has low confidence, it communicates with the human through natural-language QA. We compare against two baselines with manual trajectory selection: (1) *Human-Direct*: the robot directly queries the human with a trajectory-annotated image, and the human manually selects a preferred trajectory. (2) *Human-Assist*: the robot queries only at low confidence, and the human manually selects a preferred trajectory when queried.

The metrics used for our evaluation are as follows: *Alignment Rate* measures how often the robot’s chosen trajectory matches the human-preferred trajectory. Higher indicates more

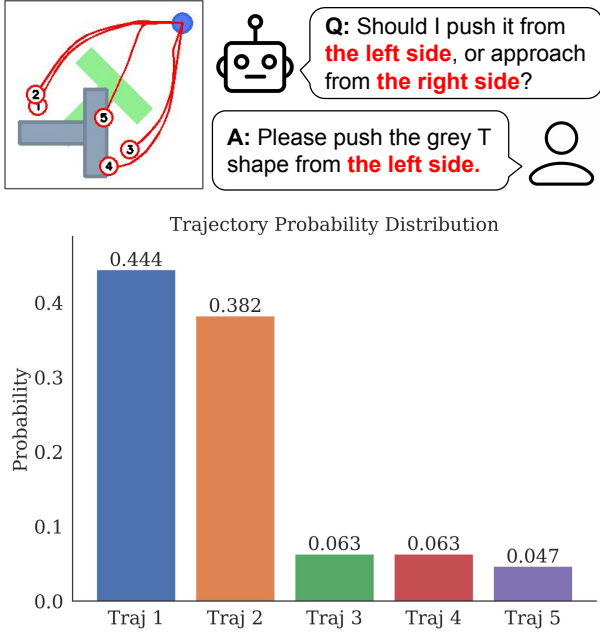


Fig. 3: Qualitative results of the VLM-generated natural-language queries. Given the top-left trajectory-annotated image, the VLM query human about their preferences. After receiving the human answer, VLM outputs a trajectory probability distribution that is aligned with human preference.

effective preference alignment. *Success Rate* measures the averaged success rate across experiment runs. This serves as a reference metric to ensure our policy steering doesn’t break task completion. *Query Times* counts how many times the robot asks the human for feedback. Smaller query times means the system understands preferences more efficiently and interrupts the user less. *Steering Times* counts how many times the system pauses to select a trajectory during execution. Smaller steering times means lower execution cost.

The results prove our first hypothesis **H1**. As shown in Figure 4, our method with VLM-based natural-language QA achieves perfect Alignment Rate and Success Rate, comparable to strong baselines with manual human selection. More importantly, our method requires fewer Query Times and Steering Times, meaning that it is significantly more efficient than baselines. By avoiding repeated manual trajectory selections, our method with VLM-based natural-language QA reduces human effort while maintaining strong performance. For each method, results are averaged over four initial configurations, with each configuration evaluated using five sample runs executed from the beginning to the end of the episode.

The example shown in Figure 5 supports our second hypothesis **H2**. The VLM confidence of human preference changes drastically in the Human-Assist baseline, while our method improves confidence and maintains it at high levels. We also extract the corresponding textual VLM reasoning. The VLM reasoning in the Human-Assist baseline shows that it struggles to get information from manual trajectory selections. In contrast, the VLM reasoning of our method confirms that

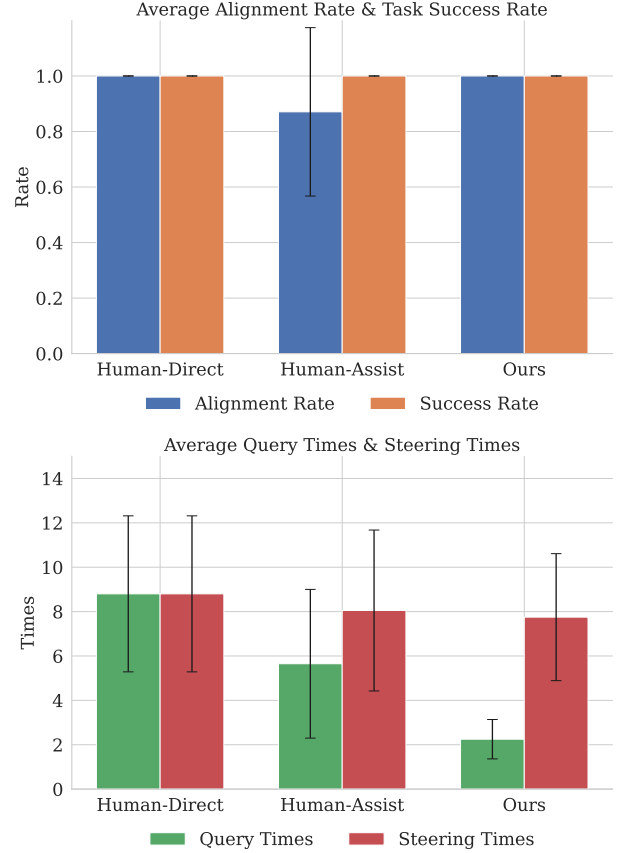


Fig. 4: Quantitative comparison between our method and two baselines (Human-Direct and Human-Assist) on average Alignment Rate ($\uparrow$), Success Rate ($\uparrow$), Query Times ($\downarrow$) and Steering Times ($\downarrow$).

our natural-language QA provides clearer signals for human preference. Overall, VLM-based natural-language QA is better at maintaining high confidence about human preference than manual trajectory selection.

D. Ablation Studies

The results shown in Figure 6 prove our 3rd hypothesis **H3**. Providing the VLM with the full conversation history context from the start of the episode significantly improves its confidence, reducing ambiguity in preference inference. As a result, the system asks fewer clarification questions, creating a more efficient interaction loop. In contrast, supplying only the last conversation context lowers confidence and causes the VLM to query the user more frequently. Empirically, the model asks about 3.8 questions with full history versus around 6 questions with only the last-turn context, and it also achieves a higher alignment rate in the full-history setting.

The results shown in Figure 7 prove our fourth hypothesis **H4**. Allowing VLM to interact with user throughout the episode makes it easier for it to infer human preference compared to the case where only initial task instruction is given, and the VLM is left alone to infer preference at every

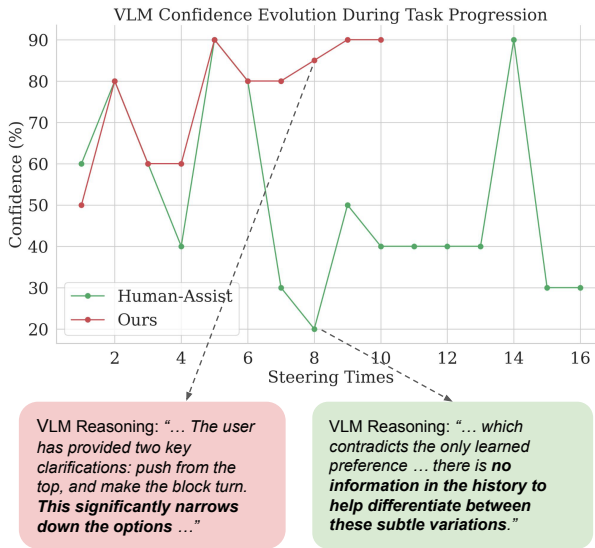


Fig. 5: VLM confidence evolution plot during task progression and the corresponding textual VLM reasoning about confidence.

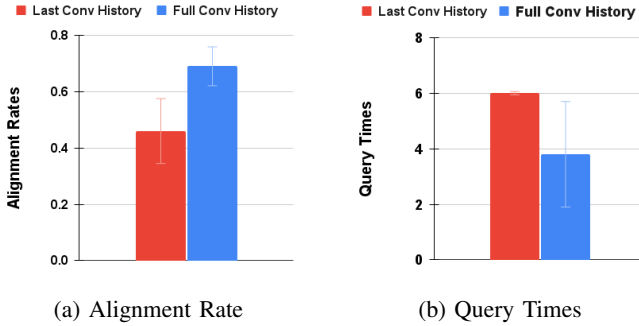


Fig. 6: Quantitative comparison between our full conversation history and last conversation history.

steering time. It is evident from the alignment rate, as it is higher in the case of continuous-user interaction, and the lower steering time indicates that the task can be finished earlier.

All the ablations are done with 2 initial configuration of the push T block and for each configuration 5 trials were ran.

VI. CONCLUSION

This work introduced a VLM-guided interactive preference alignment framework for test-time policy steering, addressing a core challenge in human-robot interaction: inferring and querying human preferences. We framed preference alignment as a probabilistic trajectory selection problem and proposed an interactive mechanism that combines visual grounding, natural-language reasoning, and confidence-aware human querying. By leveraging VLMs as preference critics, our system can evaluate multiple candidate trajectories from visual rollouts and infer human-aligned preferences directly at execution time. Crucially, instead of assuming that the VLM is always the correct evaluator, we introduced an explicit confidence estimation mechanism that determines when additional

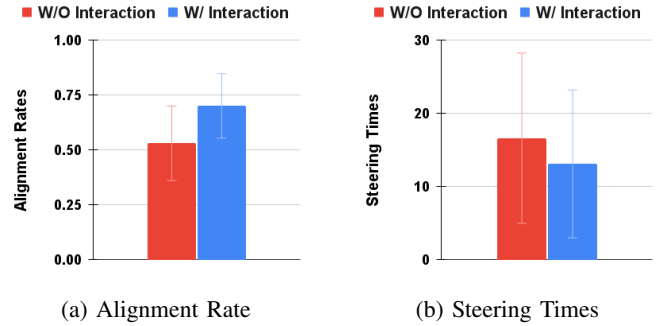


Fig. 7: Quantitative comparison between continuous user interaction and first instruction only.

human clarification is needed. This enables the robot to act autonomously when confident, while selectively querying the human only under ambiguity, reducing user mental burden.

A. Future Work

While this work demonstrates the feasibility and advantages of VLM-guided interactive preference alignment, several important directions remain open for future exploration.

First, our framework currently relies on the VLM’s self-reported confidence as a proxy for uncertainty. A more principled alternative would be to explicitly estimate uncertainty from the model’s outputs. For example, the system could prompt the VLM with each annotated trajectory using a “yes” or “no” question such as “Is this trajectory best aligned with the user’s intent?” and use the logit or probability of the “yes” token to derive both relative trajectory rankings and calibrated confidence estimates. This approach could provide a more grounded and comparable measure of preference strength across trajectories.

Second, the current implementation operates in a relatively simple 2D setting where visual differences are easily discernible. Scaling this approach to 3D environments and more complex preference structures introduces challenges in visual reasoning, spatial understanding, and inferring user intent that may not be directly observable from images alone, motivating richer multimodals.

Finally, while we focus on test-time policy steering with a fixed base policy, an important extension is to integrate online or offline learning using accumulated interaction data. Such an approach could enable long-term personalization, improve preference inference over time, and gradually adapt the underlying policy itself, bridging reactive alignment at test time with persistent user-specific behavior.

B. Contributions of Each Team Member

All members contributed to the formulation of the research idea. Diffusion policy rollouts and trajectory annotations were performed by Shohei Nagai (Push-T), Shikun Ban and Lucky Kant Nayak (Evaluations for all the hypothesis). Zixi Song developed the Human-VLM interaction interface and integrated it with the rollout pipeline. All members participated in writing and editing the report.

REFERENCES

- [1] Chris L Baker, Joshua B Tenenbaum, and Rebecca R Saxe. Goal inference as inverse planning. In *Proceedings of the annual meeting of the cognitive science society*, volume 29, 2007.
- [2] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.
- [3] Johannes Fürnkranz and Eyke Hüllermeier. Preference learning and ranking by pairwise comparison. In *Preference learning*, pages 65–82. Springer, 2010.
- [4] Udit Ghosh, Dripta S. Raychaudhuri, Jiachen Li, Konstantinos Karydis, and Amit Roy-Chowdhury. Preference vlm: Leveraging vlms for scalable preference-based reinforcement learning, 2025. URL <https://arxiv.org/abs/2502.01616>.
- [5] Kunal Handa, Yarin Gal, Ellie Pavlick, Noah Goodman, Jacob Andreas, Alex Tamkin, and Belinda Z Li. Bayesian preference elicitation with language models. *arXiv preprint arXiv:2403.05534*, 2024.
- [6] Joey Hejna and Dorsa Sadigh. Inverse preference learning: Preference-based rl without a reward function. *Advances in Neural Information Processing Systems*, 36: 18806–18827, 2023.
- [7] Borja Ibarz, Jan Leike, Tobias Pohlen, Geoffrey Irving, Shane Legg, and Dario Amodei. Reward learning from human preferences and demonstrations in atari. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/8cbe9ce23f42628c98f80fa0fac8b19a-Paper.pdf.
- [8] Mitsuhiro Nakamoto, Oier Mees, Aviral Kumar, and Sergey Levine. Steering your generalists: Improving robotic foundation models via value guidance. In *Conference on Robot Learning*, pages 4996–5013. PMLR, 2025.
- [9] Soroush Nasiriany, Fei Xia, Wenhao Yu, Ted Xiao, Jacky Liang, Ishita Dasgupta, Annie Xie, Danny Driess, Ayzaan Wahid, Zhuo Xu, et al. Pivot: Iterative visual prompting elicits actionable knowledge for vlms. In *International Conference on Machine Learning*, pages 37321–37341. PMLR, 2024.
- [10] Andi Peng, Yuying Sun, Tianmin Shu, and David Abel. Pragmatic feature preferences: Learning reward-relevant preferences from human input. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 40258–40270. PMLR, 21–27 Jul 2024. URL <https://proceedings.mlr.press/v235/peng24d.html>.
- [11] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36:53728–53741, 2023.
- [12] Allen Z Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, et al. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning*, pages 661–682. PMLR, 2023.
- [13] Allen Z. Ren, Jaden Clark, Anushri Dixit, Masha Itkina, Anirudha Majumdar, and Dorsa Sadigh. Explore until confident: Efficient exploration for embodied question answering. In *Robotics: Science and Systems (RSS)*. Robotics: Science and Systems Foundation, 2024. doi: 10.48550/arXiv.2403.15941. URL <https://arxiv.org/abs/2403.15941>.
- [14] Lucy Xiaoyang Shi, Zheyuan Hu, Tony Z Zhao, Archit Sharma, Karl Pertsch, Jianlan Luo, Sergey Levine, and Chelsea Finn. Yell at your robot: Improving on-the-fly from language corrections. *arXiv preprint arXiv:2403.12910*, 2024.
- [15] Daeun Song, Jing Liang, Xuesu Xiao, and Dinesh Manocha. VI-tgs: Trajectory generation and selection using vision language models in mapless outdoor environments. *IEEE Robotics and Automation Letters*, 10(6): 5791–5798, 2025. doi: 10.1109/LRA.2025.3559822.
- [16] Sumedh Sontakke, Jesse Zhang, Séb Arnold, Karl Pertsch, Erdem Biyik, Dorsa Sadigh, Chelsea Finn, and Laurent Itti. Roboclip: One demonstration is enough to learn robot policies. *Advances in Neural Information Processing Systems*, 36:55681–55693, 2023.
- [17] Yanwei Wang, Lirui Wang, Yilun Du, Balakumar Sundaralingam, Xuning Yang, Yu-Wei Chao, Claudia Pérez-D’Arpino, Dieter Fox, and Julie Shah. Inference-time policy steering through human interactions. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 15626–15633. IEEE, 2025.
- [18] Yufei Wang, Zhanyi Sun, Jesse Zhang, Zhou Xian, Erdem Biyik, David Held, and Zackory Erickson. RL-vm-f: Reinforcement learning from vision language foundation model feedback. In *International Conference on Machine Learning*, pages 51484–51501. PMLR, 2024.
- [19] Yilin Wu, Ran Tian, Gokul Swamy, and Andrea Baccus. From foresight to forethought: Vlm-in-the-loop policy steering via latent alignment. *arXiv preprint arXiv:2502.01828*, 2025.
- [20] Wangqi Xue, Bo An, Shuicheng Yan, and Zhongwen Xu. Reinforcement learning from diverse human preferences. In Kate Larson, editor, *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, pages 5298–5306. International Joint Conferences on Artificial Intelligence Organization, 8 2024. doi: 10.24963/ijcai.2024/586. URL <https://doi.org/10.24963/ijcai.2024/586>. Main Track.

- [21] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.

APPENDIX A PROMPTS

A. Confidence Evaluation Prompt

```
# --- ROLE: What the model should act as
---
You are an AI assistant evaluating
confidence in trajectory selection
for a robotic control task.

# --- INPUTS the model receives ---
Given:
- An annotated image showing multiple
  robot trajectories          # visual info
- A user's natural language command
                              # user intent
- Conversation history (if any)          # prior
  context

# --- CORE OBJECTIVE ---
Your task: Evaluate how confident you are
in understanding the user's
preference for trajectory selection.

# --- SAFETY / CALIBRATION ---
**IMPORTANT: Be conservative and realistic
. Do NOT be overconfident.**

# --- OUTPUT FORMAT (strict JSON schema)
---
Return a JSON response with ONLY this
structure:
{
  "confidence": <float 0.0-1.0>,          #
    uncertainty estimate
  "reasoning": "Brief explanation of
    confidence level"
}

# --- CONFIDENCE GUIDELINES ---
Confidence Guidelines (err on the side of
caution):
- 1.0: Extremely rare; only for fully
  unambiguous commands
- 0.7-0.9: High confidence but note minor
  ambiguity
- 0.4-0.6: Moderate confidence; ambiguity
  common here
- 0.0-0.3: Low confidence; ask clarifying
  questions

# --- WHEN TO LOWER CONFIDENCE ---
**Be MORE conservative if:**
- The command uses vague terms ("carefully
  ", "gently", etc.)
- Multiple trajectories could satisfy the
  command
- Important constraints are unspecified
- You must assume what the user wants
- There is ANY doubt about geometry (2D/3D
  , collisions, etc.)

# --- SELF-CHECK: Encourage explicit
```

```
reasoning ---
Consider:
- Is the command specific and measurable?
- Does conversation history provide enough
  context?
- Are multiple interpretations still
  possible?
- Would some trajectories violate intent?
- What assumptions might be wrong?
```

B. Trajectory Scoring Prompt

```
# --- ROLE: The model acts as a robotic
trajectory evaluator ---
You are evaluating trajectory options for
a robot to complete a task.

**Imagine you are controlling the robot to
fulfill the human's intent.
Which trajectory would you choose?**

# --- CRITICAL: Clarify trajectory
structure and prevent misinterpretation
---
CRITICAL - TRAJECTORY STRUCTURE (DO NOT
MISINTERPRET):
- ALL trajectories share the SAME STARTING
  POINT (robot's current position)
- The labels (Path 1, Path 2, Path 3, etc
  .) mark ENDPOINTS, not start points
- Each path = COMPLETE motion from origin
  -> labeled endpoint
- DO NOT treat indices as start points -
  they are ENDPOINTS
- Always trace the FULL PATH from origin
  to endpoint

# --- MAIN EVALUATION DIMENSIONS ---
For each candidate trajectory, assess
according to these criteria:

# (1) TASK INTENT ALIGNMENT
- Does this trajectory achieve the user's
  goal?
- Is the endpoint position suitable for
  the task?
- Does the action type match the command (
  push / grasp / position)?

# (2) PHYSICAL FEASIBILITY & INTERFERENCE
---
# Distinguish between 2D and 3D
  interpretations.
**CRITICAL:** Identify whether this is a 2
  D or 3D task.

# --- 2D planar tasks (top-down) ---
- Trajectories CANNOT overlap with planar
  objects
- ANY intersection along path = collision
  -> target disturbed
- Example: Path grazes target's side ->
  invalid endpoint
- Even minor overlaps cause unintended
  interference in 2D
```

```

# --- 3D spatial tasks (side view) ---
- Robot can move OVER or AROUND obstacles
- Consider vertical clearance and physical maneuverability
- Path may cross 2D projections safely if at different height
- Check whether 3D motion is feasible

# --- How to identify the scenario ---
- Top-down -> likely 2D planar
- Side view -> likely 3D spatial
- Use depth cues, shadows, or perspective for dimensionality

# --- Nuanced interaction handling ---
Not all overlaps are bad. Intentional contact (pushing / grasping / sliding) may be beneficial. Decide contextually whether contact is interference or purposeful interaction.

# (3) ALIGNMENT WITH GOAL STATE ---
- Does the endpoint achieve the desired final configuration?
- Is the target object correctly positioned/oriented?
- Does the outcome match the user's intent?

# (4) USEFULNESS FOR SUBSEQUENT ACTIONS ---
- Does this endpoint allow efficient follow-up actions?
- How does it affect future rollouts or strategy?
- Consider long-term utility beyond immediate success

# --- GUIDELINE REMINDER ---
These criteria are flexible aids, not hard rules.
Use reasoning and balance trade-offs per scenario.

# --- RESPONSE SCOPE CONTROL ---
User clarifications apply only to the CURRENT image unless the initial task statement specifies otherwise. Do not propagate preferences to new scenes automatically.

# --- CRITICAL: Trajectory Count Requirement ---
- Score ALL paths from Path 1 to Path N (inclusive)
- If a path is invisible or unclear, assign score = 0
- Missing trajectories are treated as 0 - better to include explicitly

# --- JSON OUTPUT FORMAT ---
Return JSON in this exact structure:
{
  "scenario_type": "2D planar" or "3D

```

```

    spatial",
    "scenario_reasoning": "Why it's 2D or 3D",
    "scores": {
      "Path 1": <scalar score>,
      "Path 2": <scalar score>,
      ...
      "Path N": <scalar score> // include all paths
    },
    "analysis": {
      "Path 1": {
        "physical_outcome": "What happens if executed",
        "collision_analysis": "Detailed collision reasoning",
        "alters_environment": true/false,
        "future_potential": "Usefulness for next steps"
      },
      "Path 2": {...},
      ...
    },
    "reasoning": "Overall explanation of scoring rationale"
  }

# --- SCORING SCALE GUIDELINES ---
- Assign scores from 0-10 per trajectory
- Avoid extremes: reserve 0-2 and 9-10 for clear failures/successes
- Most realistic paths = 3-8 range
- Reflect trade-offs between feasibility, intent, and future utility
- Nuanced scores (e.g., 5.5 vs 6.5) convey better understanding

```

C. Clarifying Question Prompt

```

# --- ROLE: The model acts as a conversational agent asking a clarifying question ---
You are asking a clarifying question to help select the best robot trajectory among multiple annotated trajectory options visible in the image.

# --- CONTEXT PROVIDED TO THE MODEL ---
Context:
- User's command: {command}
- Conversation history: {history}
- You can see multiple trajectory options but are unsure which the user prefers.

# --- PRIMARY OBJECTIVE ---
Your goal: Generate a question that helps distinguish between different trajectory options by focusing on the key aspects that differentiate them.

# --- QUESTION REQUIREMENTS ---
Generate a natural language question that:
- Focuses on CRITERIA that separate trajectories

```

(e.g., approach angle, collision avoidance, precision vs. speed, directness)

- Asks about motion or goal aspects that help choose between options
- Uses clear, conversational phrasing
- NEVER mentions path labels, colors, numbers, or annotations
(e.g., "path 1", "blue path", "trajectory 3" -> disallowed)
- Clarifies ambiguous intent aspects to narrow down which option matches best.

--- TRAJECTORY DIFFERENCE CHECKLIST ---
Think about what makes trajectories distinct:

- Do they approach from different angles or sides?
- Do they start or end from different relative positions?
- Do some avoid obstacles while others are more direct?
- Do some trade speed vs. precision?
- Do some push or interact, while others avoid contact?
- Do some terminate in different orientations or positions?
- Do they differ spatially relative to objects or boundaries?

--- SPATIAL AND DIRECTIONAL CONSIDERATIONS ---

****IMPORTANT:**** When trajectories differ spatially or directionally, ask about those distinctions.

Effective question types include:

- Relative direction (which side / from which direction)
- Spatial relationship (near, around, behind, aligned with)
- Approach angle or orientation

Such questions are most informative for clarifying spatial ambiguities.

--- EXPECTED JSON OUTPUT FORMAT ---
Return JSON:

```
{  
  "question": "Your clarifying question  
  here"  
}
```

--- EXAMPLES OF GOOD QUESTIONS ---

- "Should I approach directly or take a longer path to avoid disturbing objects?"
- "Do you want me to push it or just position next to it without contact?"
- "Do you want precise positioning or is a quick approach acceptable?"
- "Should I avoid all obstacles or is it okay to pass close to them?"
- "Do you want the final position aligned with the other block or just nearby?"

--- BAD QUESTION EXAMPLES (DO NOT ASK)

- "Should I take path 4 or path 5?" #
Mentions path numbers
- "Which trajectory should I follow?" #
Too vague
- "The blue path or the red path?" #
Mentions visual colors
- "What do you want?" #
Uninformative and broad